

AAPD7112

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Advanced Application Development

Portfolio of Evidence – Part 2

README

MuniServe – Municipal Services Application

How to Compile, Run & Use the Software

Field	Details
Module	Advanced Application Development (AAPD7112)
Project	MunicipalServicesApp1
Platform	.NET Framework 4.8 – Windows Forms (C#)
Part	Part 2 – Local Events & Announcements
Date	2026

1. Application Overview

MuniServe is a C# .NET Framework 4.8 Windows Forms desktop application that enables South African citizens to report municipal issues and browse local events. Part 2 adds the fully functional Local Events & Announcements page, built using advanced data structures.

Part 2 Features Added

- Local Events & Announcements page — now fully active on the main menu
- Keyword, category, and date search functionality
- Smart recommendation engine based on user search patterns
- Search history displayed as clickable chips (backed by a Stack)
- Urgent event banner driven by a Priority Queue
- Live statistics panel showing total events, categories, dates, and searches
- 8 pre-loaded real-world South African municipal events

Data Structures Used

Structure	Purpose
Stack	Stores the last 20 search terms — used for the Recent Searches chip strip
Queue	FIFO announcement queue — all events queued in order of addition
Priority Queue	Min-heap surfaces urgent (Priority 1) events first in the banner
SortedDictionary	Events indexed by date — auto-sorted chronologically, $O(\log n)$ lookup
Dictionary	Events indexed by category — $O(1)$ category filter lookup
Hash Table (Dict)	Records search frequency per term — drives the recommendation engine
HashSet (Set)	Tracks unique category names and unique date strings

2. System Requirements

Requirement	Details
Operating System	Windows 10 or Windows 11
.NET Framework	Version 4.8 (pre-installed on most Windows 10/11 machines)
IDE	Visual Studio 2019 or later (Community Edition is free)
RAM	Minimum 4 GB recommended
Disk Space	~50 MB for the project and build output

3. How to Compile the Application

- 1. Open Visual Studio.** Launch Visual Studio 2019 or later. Download the free Community Edition from visualstudio.microsoft.com if needed.
- 2. Open the Project.** Go to File → Open → Project/Solution. Navigate to the `MunicipalServicesApp1` folder and open `MunicipalServicesApp1.sln`.
- 3. Verify the Project File.** Right-click the project → Edit Project File and confirm it contains:
`<TargetFramework>net48</TargetFramework>` and `<UseWindowsForms>true</UseWindowsForms>`.
- 4. Add Part 2 Files.** Ensure the following files are included in the project: `Forms/LocalEventsForm.cs`, `DataStructures/EventRepository.cs`, `Models/MunicipalEvent.cs`, and the updated `Forms/MainMenuForm.cs`.
- 5. Build.** Press Ctrl + Shift + B or go to Build → Build Solution. The Output window should show: Build succeeded with 0 error(s).

4. Project Structure

File / Folder	Purpose
MunicipalServicesApp1.sln	Solution file — open this in Visual Studio
Program.cs	Application entry point
Forms/MainMenuForm.cs	Main menu — Report Issues & Local Events now both active
Forms/ReportIssuesForm.cs	Part 1 — Report Issues form (unchanged)
Forms/LocalEventsForm.cs	Part 2 — Local Events & Announcements form (new)
Models/Issue.cs	Data model for reported issues
Models/MunicipalEvent.cs	Data model for events and announcements (new)
DataStructures/EventRepository.cs	All data structures: Stack, Queue, PriorityQueue, SortedDictionary, Dictionary, HashSet (new)

5. How to Run the Application

Option A – From Visual Studio (Recommended)

- Press F5 to run with the debugger attached.
- Press Ctrl + F5 to run without the debugger.

Option B – Run the Executable Directly

- Navigate to: MunicipalServicesApp1\bin\Debug\net48\
- Double-click MunicipalServicesApp1.exe to launch.

6. How to Use the Application

6.1 Main Menu

When the application starts, the Main Menu is displayed. Three service options appear in card format. In Part 2, both Report Issues and Local Events & Announcements are now active. Service Request Status remains disabled.

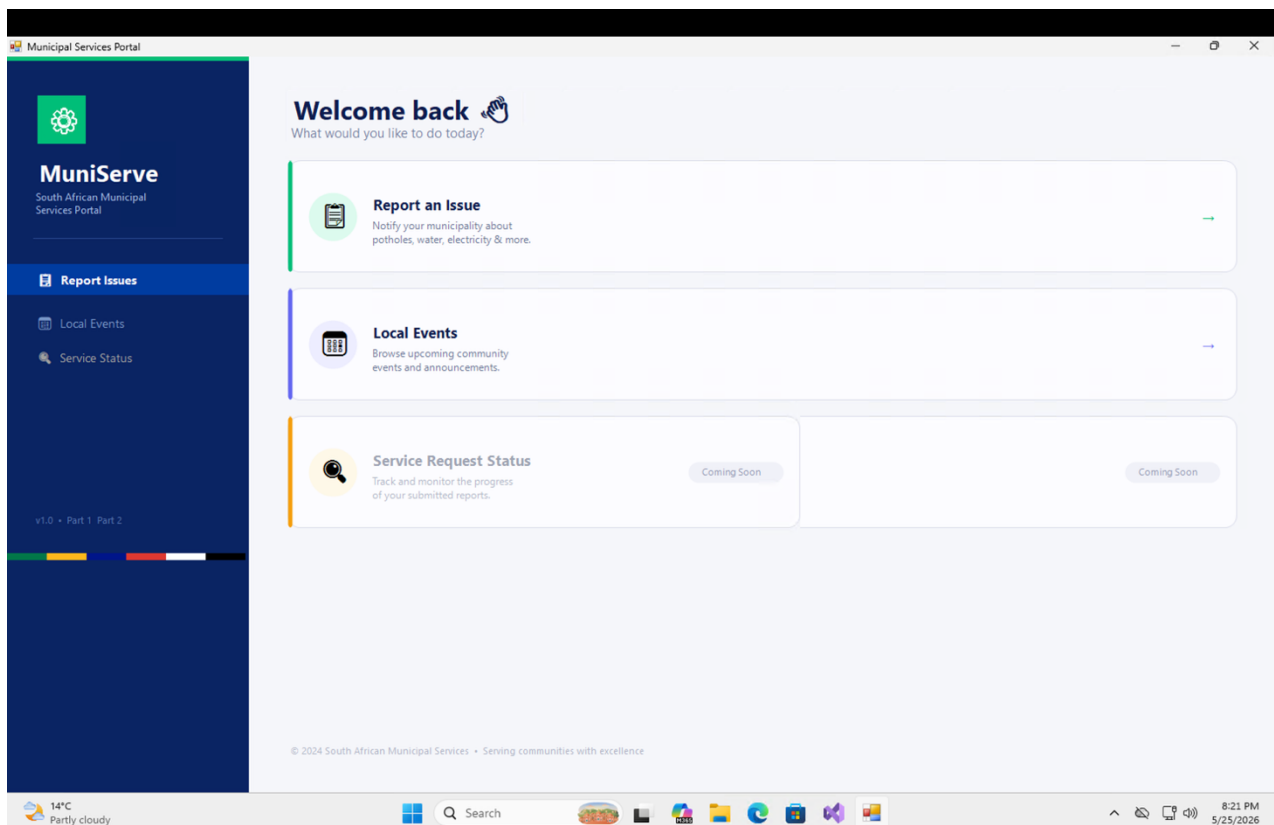


Figure 1 – Main Menu with Report Issues and Local Events & Announcements active

6.2 Local Events & Announcements Page

Click the Local Events & Announcements card to open the events page. The form loads with all 15 pre-seeded South African municipal events displayed in date order.

Local Events & Announcements

Local Events and Announcements

Recommended For You

Farmers Market
27 May 2026
Town Square

Public Safety Meeting
28 May 2026
Police Station

Community Clean-Up
30 May 2026
Central Park

Youth Sports Day
01 Jun 2026
Stadium

Title	Category	Date	Location	Description
Farmers Market	Market	27 May 2026	Town Square	Fresh local produce
Public Safety Meeting	Public Safety	28 May 2026	Police Station	Discuss community safety
Community Clean-Up	Environment	30 May 2026	Central Park	Help clean local parks
Youth Sports Day	Sports	01 Jun 2026	Stadium	Fun sports for kids
Water Conservation Workshop	Workshop	04 Jun 2026	Library Hall	Learn to save water
Recycling Drive	Environment	06 Jun 2026	Recycling Centre	Bring recyclables
Tech Expo	Workshop	09 Jun 2026	Convention Centre	Local innovation showcase
Heritage Day Celebration	Festival	14 Jun 2026	Heritage Site	Cultural event

Category: All Categories From Date: 5/25/2026 To Date: 6/25/2026 Keyword:

Search Clear

Unique Categories: 6 | Unique Event Dates: 8 (HashSet demo)

Recent searches (Queue, last 5): Cat:All Categories | 25/05-25/06 | KW:

8 event(s) found

14°C Partly cloudy

Search

8:21 PM 5/25/2026

Figure 2 – Local Events & Announcements page showing all events on load

6.3 Urgent Events Banner (Priority Queue)

At the top of the events page, a green banner displays the highest-priority event surfaced by the min-heap Priority Queue. Priority 1 events (marked Urgent) always appear here first.

The screenshot shows a web application titled "Local Events & Announcements". At the top, there is a "Recommended For You" section with four event cards:

- Farmers Market**: 27 May 2026, Town Square
- Public Safety Meeting**: 28 May 2026, Police Station
- Community Clean-Up**: 30 May 2026, Central Park
- Youth Sports Day**: 01 Jun 2026, Stadium

Below these cards is a table of events:

Title	Category	Date	Location	Description
Farmers Market	Market	27 May 2026	Town Square	Fresh local produce
Public Safety Meeting	Public Safety	28 May 2026	Police Station	Discuss community safety
Community Clean-Up	Environment	30 May 2026	Central Park	Help clean local parks
Youth Sports Day	Sports	01 Jun 2026	Stadium	Fun sports for kids
Water Conservation Workshop	Workshop	04 Jun 2026	Library Hall	Learn to save water
Recycling Drive	Environment	06 Jun 2026	Recycling Centre	Bring recyclables
Tech Expo	Workshop	09 Jun 2026	Convention Centre	Local innovation showcase
Heritage Day Celebration	Festival	14 Jun 2026	Heritage Site	Cultural event

A modal is open for the "Community Clean-Up" event, showing details: "Help clean local parks 30 May 2026 at Central Park". Below the table is a search section with filters for Category, From Date, To Date, and Keyword. The status bar at the bottom shows "14°C Partly cloudy", a search bar, and system icons.

Figure 3 – Urgent Events banner driven by the Priority Queue data structure

6.4 Search Functionality

Use the search bar to filter events by keyword, category dropdown, or specific date. The search uses a SortedDictionary for date lookups and a Dictionary for category lookups, both providing fast $O(\log n)$ and $O(1)$ performance respectively.

The screenshot displays a web application titled "Local Events and Announcements". At the top, there is a navigation bar with a "Back" button. Below this, a section titled "Recommended For You" features four event cards: "Farmers Market" (27 May 2026, Town Square), "Public Safety Meeting" (28 May 2026, Police Station), "Community Clean-Up" (30 May 2026, Central Park), and "Youth Sports Day" (01 Jun 2026, Stadium). Below the recommendations is a table with the following data:

Title	Category	Date	Location	Description
Public Safety Meeting	Public Safety	28 May 2026	Police Station	Discuss community safety

Below the table is a search filter section with the following fields:

- Category: All Categories (dropdown)
- From Date: 5/25/2026 (calendar icon)
- To Date: 6/25/2026 (calendar icon)
- Keyword: public (text input)
- Buttons: Search (green), Clear (grey)

At the bottom of the application, there is a status bar showing "Unique Categories: 6 | Unique Event Dates: 8 (HashSet demo)", "Recent searches (Queue, last 5): Cat:All Categories | 25/05-25/06 | KW:public | Cat:All Categories | 25/05-25/06 | KW:", and "1 event(s) found". The Windows taskbar at the very bottom shows the date and time as 8:22 PM on 5/25/2026.

Figure 4 – Search results filtered by keyword / category / date

The category dropdown is populated from a HashSet of unique category names. Selecting a category filters events using a Dictionary lookup for instant results.



Figure 5 – Category dropdown populated from the HashSet of unique categories

6.6 Live Statistics Panel

A statistics row below the search bar shows live counts: total events (from the repository), unique categories (HashSet size), unique dates (HashSet size), and number of searches made (hash table key count).

The screenshot displays a web interface for searching events. At the top, there is a search bar with four input fields: 'Category:' (a dropdown menu showing 'All Categories'), 'From Date:' (a date picker showing '5/25/2026'), 'To Date:' (a date picker showing '6/25/2026'), and 'Keyword:' (an empty text box). To the right of these fields are two buttons: a green 'Search' button and a grey 'Clear' button. Below the search bar, there is a large light blue rectangular area. At the bottom left of this area, there is a statistics row with the following text: 'Unique Categories: 6 | Unique Event Dates: 8 (HashSet demo)', 'Recent searches (Queue, last 5): Cat:All Categories | 25/05-25/06 | KW: | Cat:All Categories | 25/05-25/06 | KW:', and '8 event(s) found'.

Figure 6 – Live statistics panel displaying data structure sizes

6.7 Report Issues (Part 1 — Unchanged)

The Report Issues feature from Part 1 remains fully functional. Clicking Report Issues on the main menu opens the reporting form with all original functionality.

Report a Municipal Issue

— Back

Report a Municipal Issue

📍 Location
Nihakanjhaka

📁 Category
Public Safety

📝 Description
There bus drivers are reckless

📎 Attachments (optional)
+ Attach Photo or Document Air Jordan 4.jpg

Thank you for keeping your community informed! 100%

Submit Report — Clear Form

14°C Partly cloudy Search 8:26 PM 5/25/2026

Figure 7 – Report Issues form (Part 1) still fully functional in Part 2

8. Troubleshooting

Problem	Solution
Build fails with 'partial modifier' error	Delete any auto-generated LocalEventsForm.Designer.cs file Visual Studio may have created
'net48' not found	Right-click project → Edit Project File → set <TargetFramework>net48</TargetFramework>
Application won't start	Ensure .NET Framework 4.8 is installed on the machine
File locked on rebuild	Close the running .exe first (Shift+F5 in VS), then rebuild
Local Events button still greyed out	Ensure you are using the updated MainMenuForm.cs from Part 2
No events showing	Check that EventRepository.cs and MunicipalEvent.cs are both included in the project